

Impact of the Anion Structure on Coordination and Dynamics in a Localized High-Concentration Battery Electrolyte

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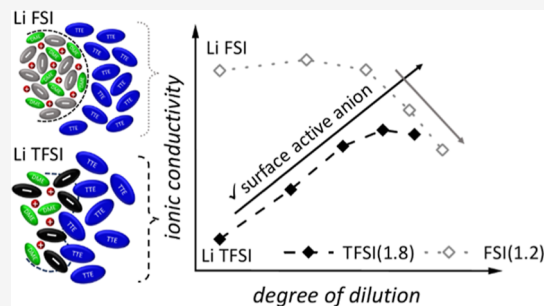
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ABSTRACT: Localized high-concentration electrolytes (LHCEs) are based on the immiscibility of a concentrated salt phase and a diluent phase, creating an internal interface. Here, we study LHCEs based on lithium bis-(fluorosulfonyl)imide or lithium bis(trifluoromethanesulfonyl)imide as the conducting salt, 1,2-dimethoxyethane as the solvent, and 1,1,2,2-tetrafluoroethyl-2,2,3,3-tetrafluoropropyl ether (TTE) as the diluent. With increasing dilution, trends of dynamic properties such as ionic conductivity and self-diffusion strongly depend on the anion structure. By analyzing ion coordinations by NMR and Raman spectroscopy, we attributed this to a different interfacial affinity of the anions: TFSI[−] anions are more likely than FSI[−] anions to interact with the diluent molecules. This stabilizes the TTE|

conducting salt interface, causing a less compact and more mobile conducting salt-rich phase. Furthermore, it leads to a higher degree of ion dissociation, which explains the enhanced molar ionic conductivity found upon dilution of TFSI-based LHCEs, as opposed to FSI-based LHCEs. These differences are most probably due to the larger size and enhanced charge delocalization of the TFSI[−] anion compared to the FSI[−] anion, making the interaction with the uncharged TTE molecule more favorable. Thus, understanding local ion coordinations in different electrolyte formulations plays a crucial role in optimizing transport properties.



INTRODUCTION

In recent years, there has been a growing interest in high-concentration electrolytes (HCEs) and, more specifically, in localized high-concentration electrolytes (LHCEs). These electrolyte classes offer the ability to enhance the electrochemical performance and safety of lithium-based batteries.^{1–5} The key feature of HCEs, which is responsible for their enhanced performance, is the increased conducting salt concentration. This leads to a high abundance of anions in the first coordination shell of the cation, which is associated with several advantages such as thermal, redox, and galvanostatic cycling stability, inhibition of lithium dendrite growth, and aluminum dissolution, resulting in an advanced performance of lithium–metal batteries (LMBs). A significant improvement during galvanostatic cycling was shown, for example, by Ren et al., who reported an 80% capacity retention after 102 cycles in a NMC811|Li battery cell using an HCE-based on lithium bis(fluorosulfonyl)imide (LiFSI) and 1,2-dimethoxyethane (DME), whereas cells containing conventional dilute electrolyte reached 80% capacity retention already after six charge/discharge cycles.⁶

However, high conducting salt concentration electrolytes come with drawbacks, like high viscosity and therefore poor ionic conductivity. Here, LHCEs present an innovative modification by maintaining the advantages of HCEs while mitigating the drawbacks through the addition of a cosolvent called diluent, which is not miscible with the salt-in-solvent phase but reduces the viscosity.^{1,2,7} Thus, two different phases

are formed: a conducting salt-rich phase consisting of the conducting salt and solvent and an independent second phase, the diluent phase. Though the size of the corresponding phases is unknown and the interface might be very fuzzy, we will in the following consider the contact line between the constituents of the salt phase with TTE as the internal interface.

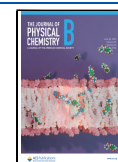
The first report of an LHCE system was published in 2013 by Dokko et al. They aimed to improve the performance of an HCE (lithium bis(trifluoromethanesulfonyl)imide (LiTFSI)/glyme) by adding a second solvent with a low donor ability, so as not to break the lithium solvation structure.⁸ By the addition of 1,1,2,2-tetrafluoroethyl-2,2,3,3-tetrafluoropropyl ether (TTE), they were able to enhance the Coulombic efficiency in a Li–S battery and further to suppress lithium polysulfide dissolution. Likewise, in the already mentioned publication by Ren et al., the performance of the cell containing an HCE system (80% capacity retention after 102 cycles) was further improved by the addition of TTE (80% capacity retention after 155 cycles).⁶ The term “localized high-concentration electro-

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lytes" (LHCEs) was introduced by Chen et al. in 2018 and is now commonly used for this type of electrolytes.⁷

Although the addition of diluent reduces the viscosity, bulk properties such as ionic conductivity and lithium transference number are not necessarily improved.^{9–12} While the ionic conductivity generally increases for small amounts of diluent due to the increasing fluidity, it decreases for a higher degree of dilution. Zhang et al. reported this behavior for the dilution of a LiFSI-2DME HCE with a variety of different diluents and showed that this maximum of ionic conductivity correlates with the fluidity of the diluent.¹³ They proposed that the initial increase in ionic conductivity is due to an additional transport mechanism (vehicular transport of ionic clusters in the diluent phase) next to ion hopping within ionic clusters, and the decreasing ionic conductivity for higher degrees of dilution is caused by the interruption of the lithium conduction in the ionic clusters. This phenomenon is also supported by other studies in the literature, which report a decrease in ionic conductivity or lithium transference number with increasing dilution.^{9,11}

Moreover, due to its prominent role in the solvation of Li⁺, the anion has a significant influence on the galvanostatic cycling performance. Lui et al.¹⁴ studied LHCEs based on sulfolane (SL), TTE, and LiFSI or LiTFSI, respectively, and observed a much higher Coulombic efficiency of >98% for a Li||Cu cell with the LiFSI-based LHCE. They attributed this to the formation of an inorganic solid–electrolyte interphase by the reduction of FSI[−] anions leading to enhanced Li-deposition/dissolution reversibility. In contrast, the SEI in the LiTFSI-based system was rich in organic species and less beneficial in terms of Li-deposition/dissolution.¹⁴

In conclusion, to overcome the drawback of interrupted cation conduction and to improve the physicochemical and electrochemical properties, an improved fundamental understanding of the complex solvation structure of Li⁺ ions in LHCE electrolytes is essential.¹⁵ In particular, the role of mutual interactions of the species, which lead to the stabilization of the internal diluent/conducting salt interface, is needed.

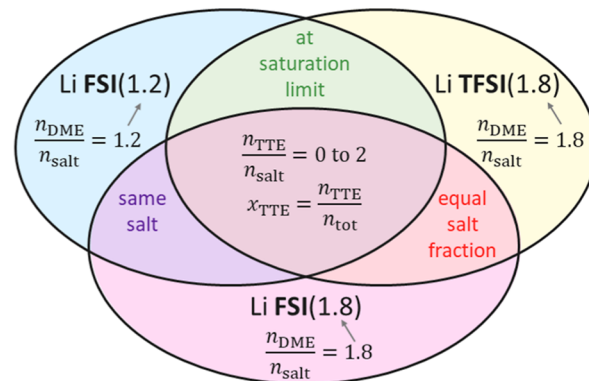
In the present work, we therefore investigate the solvation structure and more specifically the interplay between the conducting salt- and diluent-rich phases to elucidate its influence on transport properties such as ionic conductivity. To this end, we discuss LHCEs based on DME as the solvent, TTE as the diluent, and LiTFSI or LiFSI as the conducting salt, respectively. For these HCEs and LHCEs, we investigated the influence of an increasing amount of diluent. The coordination structure of the ions, defined by the direct environment of the coordinating molecules or ions, is subsequently modified. Elucidating this coordination structure in dependence on the type of anion by evaluating nuclear magnetic resonance (NMR) chemical shifts, spin relaxation data, and Raman spectra, we provide a structural explanation for the differences we observed in the dynamic properties in dependence on the anion structure.

MATERIALS AND METHODS

Materials and Electrolyte Preparation. Both conducting salts LiFSI (TCI, >98%) and LiTFSI (Sigma-Aldrich, 99%) were dried at 100 °C under high vacuum (10^{−6} hPa) overnight. The solvent DME (Sigma-Aldrich, >99.5%, anhydric) and diluent 1,1,2,2-tetrafluoroethyl-2,2,3,3-tetrafluoropropyl ether (TTE, Merck, >99%) were dried using a molecular sieve. All

materials were stored and handled in an argon-filled glovebox. The electrolytes were prepared by dissolving the lithium salt in DME and adding the respective amount of TTE. According to Scheme 1 and Table S1, a total of 15 electrolytes were

Scheme 1. Schematic Description of the Molar Composition of the Three Electrolyte Formulations^a



^aFor all systems, the molar fraction of TTE increases from 0 to 2 in relation to the lithium salt, and the molar fraction of DME is 1.2 for the LiFSI(1.2) formulation and 1.8 for the LiFSI(1.8) and LiTFSI(1.8) electrolytes. x_{TTE} defines the molar ratio of the diluent in relation to the entire electrolyte. Details of molar compositions see Supporting Information, Table S1.

formulated in this way. The rationale was to compare on the one hand two electrolyte series at the respective salt solubility limit of the HCE, yielding LiFSI(1.2), which describes LiFSI/DME with molar fractions of 1:1.2, and LiTFSI(1.8), describing LiTFSI/DME 1:1.8. On the other hand, involving a series LiFSI(1.8) provides a comparison to LiTFSI(1.8) at the same molar salt ratio. Further parameters characterizing the compositions, such as the conducting salt concentration and the oxygen-to-lithium ratio are listed in Table S2.

NMR Spectroscopy, Spin Relaxation, and Pulsed Field Gradient (PFG) NMR Diffusion. Nuclear magnetic resonance (NMR) measurements were performed with a Magritek Spinsolve spectrometer operating at a 60 MHz ¹H frequency and providing magnetic field gradients up to 0.16 T/m. Experiments were conducted at 299.65 K for the nuclei ¹H, ⁷Li, and ¹⁹F. Self-diffusion coefficients, which represent the mean square displacement of a molecule in the absence of a chemical potential gradient, were determined via pulsed field gradient NMR using the stimulated echo sequence with gradient pulses.¹⁶ Longitudinal and transverse spin relaxation rates, R_1 and R_2 of ¹⁹F and ⁷Li, were measured using the inversion recovery and the Carr–Purcell–Meiboom–Gill experiment, respectively.^{17–19} For better resolution, a 400 MHz Bruker AVANCE II HD spectrometer was used to perform single-pulse experiments to analyze chemical shifts. The ¹H chemical shift of the TTE molecule was measured against an external reference using a Teflon capillary filled with acetone. Beforehand, the chemical shift of the acetone reference was measured against a tetramethylsilane reference and determined as 2.15 ppm.

Raman Spectroscopy. To obtain Raman shifts, a MultiRAM FT-Raman spectrometer (Bruker, Ettlingen) equipped with a 1064 nm laser with a power of 264 mW was used. All samples, contained in an NMR tube, were measured in the range of 30–3600 cm^{−1} with a resolution of 1

cm^{-1} . In the case of LiTFSI-containing samples, software OPUS 8.5 (Bruker) was used to deconvolute the vibrational band of the TFSI[−] anion, identifying two peaks ($\sim 743\text{ cm}^{-1}$ and $\sim 748\text{ cm}^{-1}$), representing free and coordinated anions, respectively, by using line shapes of 40% Gauss and 60% Lorenz.

Impedance Spectroscopy. 750 μL of each electrolyte was filled in in-house build conductivity cells, which have been described elsewhere²⁰ and which were sealed under an argon atmosphere. These cells were placed in a temperature chamber (Mettler TTC256, 0.1 K temperature setting accuracy), and the impedance measurements were performed on a Metrohm Autolab/M204 potentiostat/galvanostat with 12 channels and 8-channel multiplexer for a total of 96 channels in the range of 50 Hz to 1 MHz using in-house developed electrodes. The measuring temperature of 299.15 K was held for 2 h for equilibration before the measurement.²⁰ The acquired data were fitted with an equivalent circuit, consisting of a series of a resistor and a parallel element of resistor and constant phase element, to obtain the electrolyte resistance, using software ZView Version 3.0 (Scribner Associates). The ionic conductivity was obtained as the quotient of the cell constant, previously determined using an aqueous KCl standard solution and the electrolyte resistance.

Viscosity Measurements. The viscosities of the electrolyte formulations were measured with an automatic kinematic viscosimeter (SVM 3001, Anton Paar, modified Couette measuring principle). A sample volume of 1.5 mL was inserted using a syringe, and measurements of the dynamic viscosity were performed at 299.65 K. After each measurement, the instrument was purged with acetone and dried with the integrated fan.

Density Measurements. Density measurements were performed at 299.65 K by using a DDM 2910 automatic density meter (Rudolph Research Analyzer). Before each measurement, the instrument was purged with acetone and dried with dry nitrogen. Afterward, a sample volume of 1 mL was inserted with a syringe, and the measurement was carried out.

RESULTS AND DISCUSSION

Sample Selection. In this study, we compare three different HCE and LHCE formulations (see Scheme 1): first, two HCEs near the saturation limit based on the conducting salts LiTFSI and LiFSI, respectively, are investigated. In both cases, the saturation limit was determined by incrementally increasing the conducting salt content until no more salt was soluble in DME. Thereby, a molar conducting salt to solvent ratio of 1 to 1.8 in the LiTFSI-based system and a slightly higher molar ratio of 1 to 1.2 for the LiFSI-based electrolyte were identified. Afterward, an increasing molar fraction of TTE (0 to 2) was added to obtain LHCEs with a rising degree of dilution. In the following, these two electrolyte formulations are referred to as LiTFSI(1.8) and LiFSI(1.2), respectively. Additionally, to compare both systems at the same molar ratios, part of the experiments was extended to a LiFSI-based LHCE with the same conducting salt to solvent ratio as the LiTFSI-based system, referred to as LiFSI(1.8).

Viscosity and Ionic Conductivity. As mentioned in the Introduction section, the main goal of LHCEs is to achieve a higher ionic conductivity by reducing the viscosity with the addition of a diluent. These two quantities are shown in Figure 1a in dependence on the amount of diluents in different LHCE

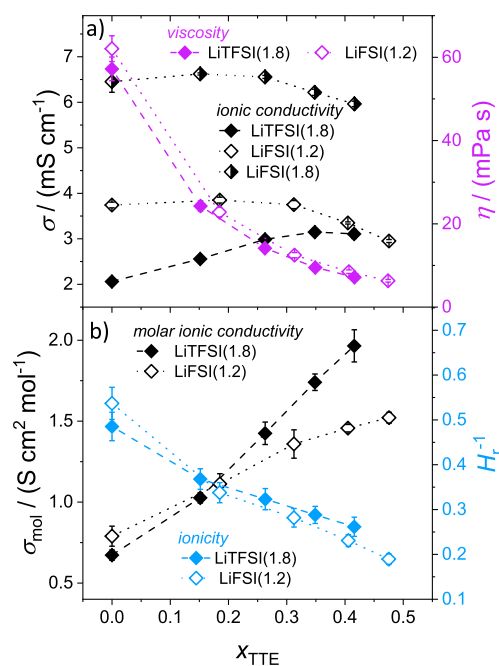


Figure 1. (a) Viscosity, η , (purple) for LiFSI(1.2) and LiTFSI(1.8) in dependence on the molar fraction of diluent x_{TTE} (open and filled violet diamonds, respectively). Ionic conductivities, σ , (black) for these two systems and for LiFSI(1.8) (half-filled diamonds). (b) Molar conductivities, σ_{mol} , (black) as well as inverse Haven ratio, H_T^{-1} , (light blue) with increasing dilution for LiFSI(1.2) (open symbols) and LiTFSI(1.8) (filled symbols).

compositions. The viscosities (purple data points) show a drastic decrease by a factor of almost ten due to the dilution by TTE. Interestingly, the absolute values are similar for the two electrolyte formulations, although the conducting salt concentration is higher in the LiFSI-based system. In general, quantitatively similar effects of dilution on the electrolyte viscosity (reduction from $>100\text{ mPa s}$ to $\sim 5\text{ mPa s}$) have been reported and are widely known for other LHCE systems.^{10,11,21–23}

The ionic conductivity of the LiTFSI-based electrolyte is in general lower than that of both LiFSI-based systems (see Figure 1, filled versus open and half-filled symbols); however, σ of LiTFSI(1.8) increases with dilution, while σ decreases for both LiFSI-based formulations. Moreover, we observe $\text{LiFSI(1.8)} \gg \text{LiFSI(1.2)} > \text{LiTFSI(1.8)}$ over the whole investigated x_{TTE} range. The significantly higher absolute values for σ in the LiFSI(1.8) electrolyte system can be explained by generally increased dynamics for these HCE and LHCEs with a conducting salt concentration further below the saturation limit. Moreover, a possible explanation for the slightly higher ionic conductivities in the LiFSI(1.2) system compared to the LiTFSI-based electrolyte formulation, which have the same viscosity, is the higher conducting salt concentration, i.e., charge carrier density, in this system. As already mentioned, we observed different trends for σ (increasing vs. decreasing with x_{TTE}) depending on the anion involved. This indicates that growing TTE phases on the one hand restrain the ion motion and lower the ionic conductivity in the LiFSI-based electrolytes regardless of the conducting salt concentration but on the other hand enhance the charge transport in the LiTFSI-based system. This different behavior upon dilution depending on the anion has been reported for

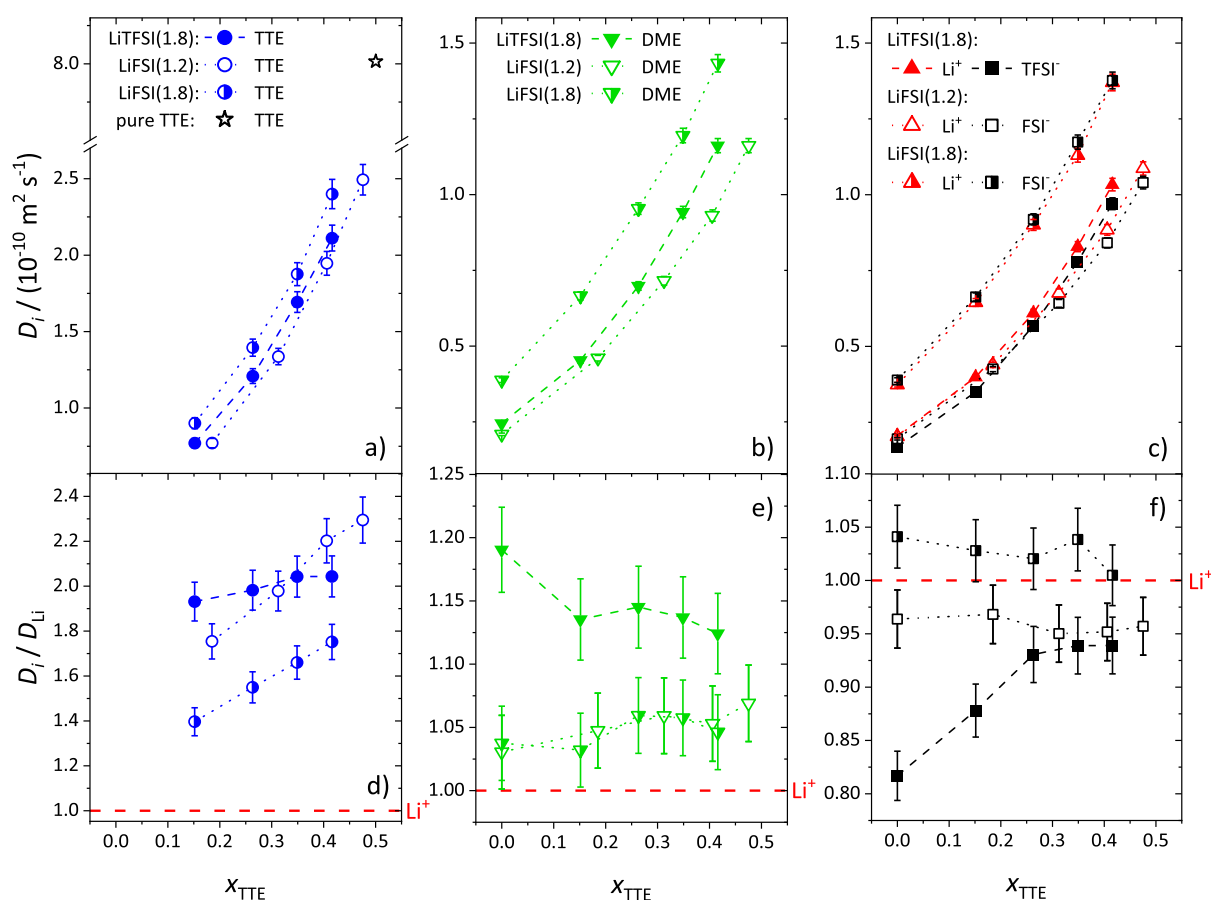


Figure 2. Self-diffusion coefficients D_i of (a) TTE (blue circles), (b) DME (green triangles) ($D_{\text{pureDME}} = 2.8 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$), and (c) Li^+ (red triangles) and anion (black squares). Self-diffusion coefficient ratios D_i/D_{Li} for (c) TTE (blue circles), (d) DME (green triangles), and (e) the anion (black squares) in dependence on the molar fraction of diluent, x_{TTE} . The systems LiTFSI(1.8), LiFSI(1.2), and LiFSI(1.8) are depicted with filled, open, and half-filled symbols, respectively.

other LHCE systems.^{11,21,23} In particular Wang et al. observed the same contrary behavior (decreasing σ versus increasing σ), when comparing LiFSI- and LiTFSI-based LHCEs using dimethyl carbonate as the solvent instead of DME.²⁴

Similar conclusions can be drawn from the molar ionic conductivities, σ_{mol} shown in Figure 1b. The molar ionic conductivities are obtained from the molar conducting salt concentrations with regard to the entire LHCE, as calculated from the electrolyte densities (Table S2). Figure 1b depicts that σ_{mol} increases more strongly for the LiTFSI-based system, which is in line with the increasing trend of σ .

Self-Diffusion. Figure 2a–c shows the self-diffusion coefficients D_i , measured via PFG-NMR spectroscopy, of TTE, DME, and the ions, respectively. Our color scheme for the different constituents of the electrolyte, which we keep throughout the paper, is blue for diluent, green for solvent, red for Li^+ , and black for the anions. We keep the notation of full symbols for LiTFSI(1.8), open symbols for LiFSI(1.2), and half-filled symbols for LiFSI(1.8). As expected, all self-diffusion coefficients increase with dilution, in line with decreasing viscosity. Moreover, D_{TTE} is considerably higher compared to the self-diffusion coefficients of the remaining components (see different scales of Figure 2a,b/c). This effect is known from literature for other LHCEs and emphasizes that TTE forms a separate diluent-rich phase mostly independent of the conducting salt phase.^{9–11} This does not apply for the solvent DME. Comparing Figure 2b,c shows for all systems that the

diffusion coefficient of DME (marked in green) is comparable to those of the ions (red and black) and, further, significantly lower compared to pure DME ($D_{\text{pureDME}} = 2.8 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$). Therefore, this implies that DME is part of the conducting salt phase coordinating Li^+ , which leads to slower self-diffusion.

Concerning the self-diffusion of the ions, Figure 2c shows almost equal self-diffusion coefficients for cations (red) and anions (black) in each system, respectively, hinting at strong cation–anion coordination and correlated motion. Moreover, the absolute values of the self-diffusion coefficients are comparable for the systems LiFSI(1.2) and LiTFSI(1.8), which goes along with the similar viscosities (see Figure 1a), while D_i is generally higher in the less concentrated LiFSI(1.8) system due to increased system dynamics.

Additionally, even if TTE seems to form a separate phase, the black star in Figure 2a reveals that the diffusion of TTE in the LHCE is still much lower compared to pure TTE. Due to the presence of the salt-rich phase, obstruction effects occur, reducing the self-diffusion of TTE.

Moreover, in Figure 2d, D_{TTE} normalized to the diffusion coefficient of Li^+ , D_{Li} , is shown for the different electrolyte formulations. This ratio is a measure of how dilution influences the TTE phase compared to the conducting salt phase and reveals small differences between the LiTFSI- and LiFSI-based systems. For both LiFSI-based systems, the ratio $D_{\text{TTE}}/D_{\text{Li}}$ increases by dilution with TTE, while the ratio remains constant for LiTFSI(1.8). Thus, the diffusion in the TTE

phases of the LiFSI-based systems increases more strongly with increasing x_{TTE} than that in the conducting salt phase. This finding is consistent with the decreasing trend of σ . In contrast, in the LiTFSI-based system, the $D_{\text{TTE}}/D_{\text{Li}}$ -ratio is constant; thus, the self-diffusion coefficients of TTE and Li^+ increase equally, demonstrating an identical boosting effect of TTE addition on transport in both the TTE phase and the conducting salt-rich phase. This goes along with an increasing σ . These distinct differences in the effect of dilution on transport in either phase point to a strong influence of the type of anion on the mesoscale structure of the phases, which will be discussed further below.

This differing behavior of LiFSI- and LiTFSI-based LHCEs is supported by the inverse Haven ratio, H_{R}^{-1} , also known as ionicity, which can be calculated by using the self-diffusion coefficients of the ionic species. It is defined as the ratio of the ionic conductivities determined by impedance spectroscopy, σ , and estimated from PFG-NMR diffusion, σ_{NE} , see eqs 1 and 2. Here, F represents the Faraday constant ($96,485 \text{ s A mol}^{-1}$), R represents the gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$), T is the temperature (299.65 K), and z and c ($1.5\text{--}5 \text{ mol L}^{-1}$) are the charge number and the concentration of the ionic species i , respectively.

$$H_{\text{R}}^{-1} = \frac{\sigma}{\sigma_{\text{NE}}} \quad (1)$$

$$\sigma_{\text{NE}} = \left(\frac{F^2}{RT} \right) \sum_i z_i^2 c_i D_i \quad (2)$$

H_{R}^{-1} is shown in Figure 1b and decreases with x_{TTE} for both systems, indicating that σ_{NE} increases compared to σ . Thus, diffusive processes are more strongly accelerated than ionic conduction. While the addition of the diluent enhances the diffusion coefficients of both ion constituents, it does not promote charge transport by the same factor. This is evidence for the strongly correlated transport of cations and anions and provides again an indication of the persistence of a conducting salt phase upon dilution. Further, the ionicity decrease is more pronounced in the LiFSI-based system, which is in line with the just discussed increasing $D_{\text{TTE}}/D_{\text{Li}}$ ratio: as the Haven ratio reflects positive ion–ion correlations, these appear to play a larger role in the FSI-based LHCE, which is also in line with the less enhanced diffusivities compared to D_{TTE} and the reduced molar conductivity of this system.

Additionally, electrolyte transport properties are often demonstrated in a Walden plot, which is shown in the Supporting Information, Figure S1. Upon dilution, the fluidities, η^{-1} , as well as the molar ionic conductivities, σ_{mol} , are increasing for both series LiTFSI(1.8) and LiFSI(1.2). This trend is in line with the finding in a LiFSI/SL-based LHCE.¹⁰ Interestingly, the slope in the Walden plot is generally lower than that of the ideal KCl line, which is in accordance with the ionicity decreasing upon dilution. Thus, mass transport is enhanced by a larger factor than charge transport due to the correlated transport of cations and anions described above. Interestingly, the slope of the LiFSI-based LHCE is lower, which reflects the difference in ionicity, see discussion above, and confirms a larger contribution of positive ion–ion correlations in case of LiFSI(1.2).

Distinct differences in the development of the self-diffusion coefficients of the components building up the conducting salt phase can be discussed on the basis of Figure 2e,f, showing the

self-diffusion coefficients of DME and the anion, again normalized on D_{Li} , respectively. First of all, in the LiTFSI-based system we generally observe an order of the self-diffusion coefficient ratios of $\text{DME} > \text{Li}^+ > \text{TFSI}^-$, most pronounced in the HCE and the only slightly diluted compositions. Thus, DME diffuses on average faster and the anion is slower with regard to the cation. This might be explained by an effect, which we recently demonstrated for another highly concentrated solvent-in-salt electrolyte:²⁵ here, a heterogeneous Li^+ ion coordination leads to a faster and a slower Li-species, i.e., heterogeneous lithium diffusion. Li^+ ion is either partially coordinated by a bidentate coordination of DME molecules or trapped in a coordination governed by anions. Since TFSI^- anions can coordinate more than one Li^+ ion at the same time, cations in this coordination environment are less mobile compared to Li^+ ions partially coordinated by DME, leading to a slower species. In contrast to the anion, due to its bidentate coordination behavior, DME coordinates only one cation leading to a faster diffusing Li^+ species. This behavior was observed for low oxygen-to-Li ratios of $\text{O}_{\text{tot}}/\text{Li}^+ < 10$ and especially for a strong undersaturation of Li^+ by DME ($\text{O}_{\text{DME}}/\text{Li}^+ < 2$). Assuming that, in this case, almost all DME molecules are part of the first lithium coordination shells and thus of a fast diffusing species, while the Li^+ diffusion is averaged over fast and slow species, this leads to an order of diffusivities following $D_{\text{DME}} > D_{\text{Li}} > D_{\text{TFSI}^-}$. In analogy to these previous findings²⁵ and with regard to Figure 2e,f, this order of self-diffusion coefficients is present over the entire dilution range for the LiTFSI-based LHCEs ($\text{O}_{\text{DME}}/\text{Li}^+ = 3.6$ and $\text{O}_{\text{salt+DME}}/\text{Li}^+ = 7.6$). Even though the $\text{O}_{\text{DME}}/\text{Li}^+$ ratio is >2 here, it is still below the saturation limit of five to six oxygen atoms per Li^+ cation. Therefore, the heterogeneity in lithium-ion coordination can account for the observed order of diffusivities even in LHCE.

Interestingly, though the order of $D_{\text{DME}} > D_{\text{Li}} > D_{\text{anion}}$ is still valid in the LiFSI(1.2) system ($\text{O}_{\text{DME}}/\text{Li}^+ = 2.4$ and $\text{O}_{\text{salt+DME}}/\text{Li}^+ = 6.4$), the differences in the diffusion coefficients are less pronounced. Furthermore, the order of anion and cation diffusivity is inverted in the less concentrated LiFSI(1.8) electrolyte ($\text{O}_{\text{DME}}/\text{Li}^+ = 3.6$ and $\text{O}_{\text{salt+DME}}/\text{Li}^+ = 7.6$) (see the half-filled symbols in Figure 2f). It appears that the heterogeneity of the Li^+ ion coordination impacts diffusion less in the case of LiFSI-based electrolytes. Since the O/Li^+ ratio in the LiFSI(1.8) LHCE equals the O/Li^+ ratio in the LiTFSI(1.8) system and the cation is even more under-coordinated in the LiFSI(1.2) electrolyte formulation, this cannot be explained by a change in the conducting salt-to-solvent ratio. Thus, rather the different size of the anion may be important, influencing the structure of the salt-rich phase, which will be discussed in the following section.

Furthermore, the self-diffusion coefficient ratios of the constituents in the salt phase of both LiFSI-based systems are rather constant and close to one (see the open and half-filled symbols in Figure 2e,f). Thus, all three molecules are equally affected by dilution independent of the conducting salt concentration. Therefore, it seems that neither composition nor coordination structure of the conducting salt-rich and diluent phase change significantly with dilution, but diffusivity generally increases. In contrast, this is not the case in the LiTFSI-based system. Here, $D_{\text{TFSI}}/D_{\text{Li}}$ shows a significant increase upon diluting with TTE (see the black full squares in Figure 2f). Therefore, the growing TTE phases and TTEI conducting salt interfaces apparently have a larger positive

effect on the anion diffusion as compared to cation diffusion, which implies a change of the coordination in the conducting salt phase. This interesting finding will be rationalized further below by a reduction of the Li^+ -coordination of TFSI^- observed in Raman spectroscopy.

Composition of the Diluent Phase. Since the chemical shift of a nucleus is sensitive to its electronic environment, it can be used as a basis for interpreting the coordination structure. First of all, in order to clarify a potential modification of the molecular composition of the TTE phase, i.e., by partitioning of DME into this phase, the ^1H chemical shifts of the diluent, δ_{TTE} , in the LHCE are compared with the chemical shift in pure TTE and in binary TTE/DME mixtures. The ^1H NMR spectra of the TTE/DME mixtures are depicted in Figure S2. These reference spectra served to distinguish whether the diluent phase consists entirely of TTE molecules or if a small fraction of DME is dissolved in the TTE phase. The comparison with the ^1H chemical shift of TTE with this reference mixtures is shown in Figure 3 for one of the protons,

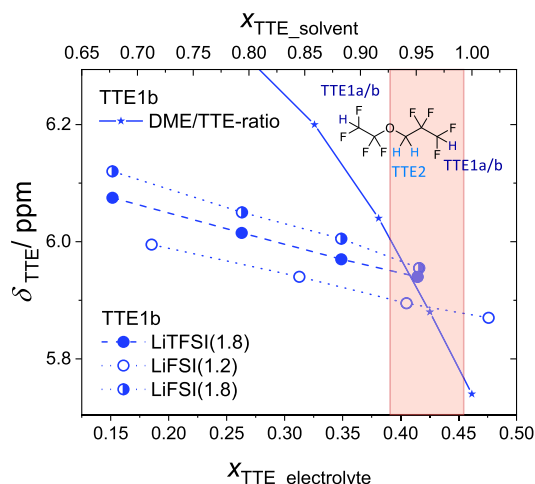


Figure 3. ^1H chemical shift, δ_{TTE} , of one (TTE1b) of the three chemically nonequivalent protons in the three electrolyte formulations LiTFSI(1.8), LiFSI(1.2), and LiFSI(1.8) in dependence on the molar fraction of TTE, $x_{\text{TTE, electrolyte}}$ (filled, open, and half-filled circles, respectively; lower x-axis). ^1H chemical shift of TTE1b in pure TTE ($x_{\text{TTE, solvent}} = 1.00$) and DME/TTE-mixtures ($x_{\text{TTE, solvent}} < 1.00$) (stars, upper x-axis). The full comparison for all three nonequivalent protons is shown in Figure S3.

exemplarily. The full comparison of all three nonequivalent protons is shown in Figure S3, exhibiting similar trends for all ^1H chemical shifts. Figure 3 shows that δ_{TTE} decreases significantly when the TTE content in the DME/TTE mixture is increased (see stars and solid lines). Thus, δ_{TTE} is sensitive to a small fraction of DME in its coordination environment, leading to a deshielding of the TTE protons. However, the chemical shift of the proton TTE1b of TTE in the LHCEs remains quite constant regardless of dilution or anion type (see circles and dashed or dotted lines). This indicates that the composition of the diluent phase is comparable between the systems and relatively constant with dilution. By comparing the absolute values of the chemical shifts in the electrolyte to the shifts obtained from the DME/TTE series, we found $\delta_{\text{TTE}}(\text{LHCE}) \approx \delta_{\text{TTE}}(95:5 \text{ TTE/DME-mixture})$ for all protons (see also Figure S3). Thus, it can be concluded that even though DME and TTE are miscible, the TTE phase in all

LHCE formulations mostly consists of TTE molecules since $\sim 95\%$ of all DME molecules remain coordinated to cations in the highly concentrated conducting salt phase. This result proves the assumption of structural heterogeneity, which is the basis of the concept of LHCEs.

Even if no huge differences are observed for the different LHCE systems, we note that δ_{TTE} is always slightly lower in the higher concentrated LiFSI(1.2) system, and thus, it is closer to the chemical shift of pure TTE. In conclusion, a higher amount of DME might partition into the TTE phase for the LiFSI(1.8) system compared to the LiFSI(1.2) formulation, consistent with the larger share of DME in the LiFSI(1.8) LHCE. However, the composition analysis based on TTE shifts has to be taken with some caution as another possible factor influencing δ_{TTE} is interactions of TTE molecules with the species in the conducting salt phase at the interface of these two domains. We therefore remain with the conclusion of almost all DME being present in the salt-rich phase and refrain from analyzing further differences of x_{DME} in TTE.

Coordination of Anions. After the molecular composition of the diluent phase is elucidated, Raman measurements can provide a deeper understanding of the coordination structure in the conducting salt phase. In the following, we analyze the S–N–S stretching vibration of the anion, which is sensitive to changes in the Li^+ coordination.¹⁵ Figure 4a–c shows the changes in this vibrational band with an increasing amount of diluent in the LiFSI(1.2), LiFSI(1.8), and LiTFSI(1.8) electrolyte formulations, respectively. Comparing the stretching vibrational bands of the FSI^- (Figure 4a,b) and TFSI^- anion (Figure 4c) shows a much broader band for the FSI^- . The narrower vibrational band of TFSI^- allows for deconvolution by two bands: a stronger band around 748 cm^{-1} decreasing with dilution and a smaller band at lower wavenumbers around 743 cm^{-1} increasing with TTE addition. In accordance to literature, these bands can be assigned to coordinated and free TFSI^- anions, respectively.^{26,27} In the spectrum of a low-concentrated 1 M LiTFSI/DME electrolyte (light blue line in Figure 4c), these two Raman bands are observed as well; however, the Raman bands are slightly shifted to lower wavenumbers. A representative deconvolution of the vibrational band is shown in Figure S4a. As expected, free anions are the dominant species in the low-concentrated 1 M LiTFSI/DME electrolyte (77%, as extracted from Figure 4c), while the fraction of free anions is generally less than 49% in the high-concentrated LHCE. Here, the fraction of free anions increases linearly from 43% to 49% with increasing dilution (Figure S4b), which indicates an increasing degree of ion dissociation upon TTE addition. This trend is in line with the increase in the $D_{\text{TFSI}}/D_{\text{Li}}$ -ratio, as observed in Figure 2f, since TFSI^- anions released from Li coordination can exhibit faster diffusion. The enhanced dissociation can be explained by anions interacting with TTE molecules rather than Li^+ ions, thus stabilizing the growing TTE|conducting salt interface instead of coordinating Li^+ .

As already mentioned, due to the relatively broad vibrational band of the FSI^- anion, it is difficult to deconvolute this band with regard to the contribution of the different fractions of anions coordinating none, one, or more cations. In accordance to the work of Han et al. studying a LiFSI/acetonitrile mixture over a broad range of salt concentrations,²⁸ we marked four Raman bands in Figure 4a,b (dotted vertical lines): two bands corresponding to free FSI^- anions (720 cm^{-1} and 728 cm^{-1}), one band at 741 cm^{-1} attributed to Li^+/FSI^- contact ion pairs

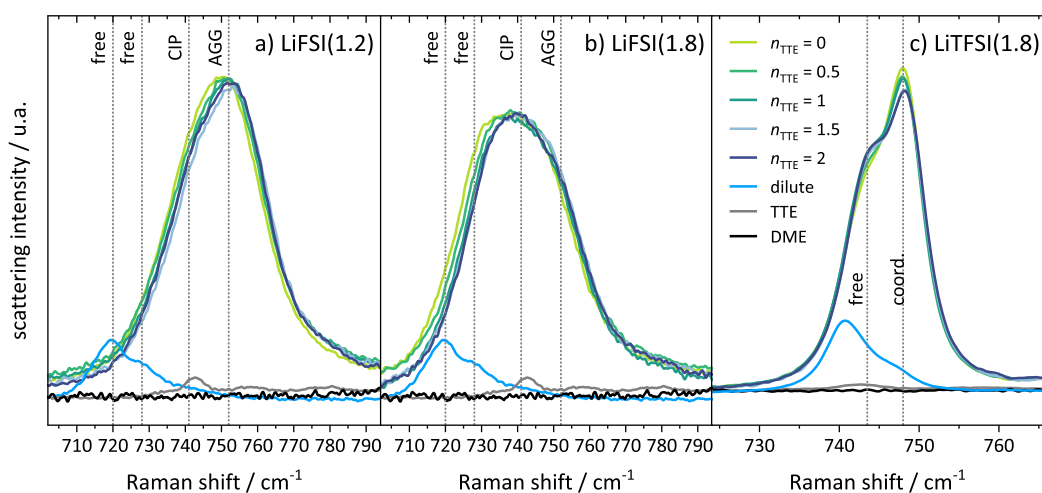


Figure 4. Raman spectra of the electrolyte formulations of (a) LiFSI(1.2), (b) LiFSI(1.8), and (c) LiTFSI(1.8) with an increasing amount of TTE (from green to dark blue). Additionally, Raman spectra of pure DME (black), pure TTE (gray), an 1 M LiFSI in DME electrolyte formulation (light blue in a and b), and an 1 M LiTFSI in DME electrolyte formulation (light blue in c).

(CIP), and one band at 752 cm^{-1} assigned to aggregated anions coordinating more than one cation.

A comparison of the LHCE Raman spectra with that of a low-concentration 1 M LiFSI/DME electrolyte (light blue line) clearly shows that almost all anions are coordinated to at least one Li^+ cation in the LHCEs, whereas free FSI^- anions are predominant in the 1 M electrolyte formulation. However, differences are observable when comparing the band shape in the higher and lower concentrated LiFSI-based LHCEs: while the contribution of aggregated anions dominates over the contribution of CIPs in the LiFSI(1.2) system (Figure 4a), it is the opposite in the LiFSI(1.8) electrolyte formulation (Figure 4b). This difference can be explained by a higher Li–DME coordination number due to the higher amount of DME in the LiFSI(1.8) system, resulting in a smaller Li–FSI coordination number.

Additionally, Figure 4a,b shows a slight blue-shift of the vibrational band of FSI^- with dilution. A similar behavior was reported by Watanabe et al.¹⁰ for the dilution of a LiFSI/SL HCE with TTE. They attributed this blue-shift to a “fragmentation of the $\cdots\text{Li}^+\cdots\text{SL}\cdots\text{Li}^+\cdots$ network creating isolated ionic clusters”. They argued that “this structural change strengthens the electrostatic $\text{Li}^+\cdots\text{SL}$ and $\text{Li}^+\cdots\text{FSA}^-$ interactions within the clusters”. It seems that a comparable effect is observed here. The growing TTE phase causes enhanced ionic interactions and a more compact conducting salt-rich phase, which is consistent with the previously discussed decreasing ionic conductivity, low H_R^{-1} , and increasing $D_{\text{TTE}}/D_{\text{Li}}$ -ratio. In contrast to the LiFSI-based LHCEs, no blue-shift of the stretching mode of TFSI^- is observed with an increasing amount of the diluent. Again, this is in accordance with literature: Liu et al. reported the already discussed blue-shift for a LiFSI/SL/TTE LHCE and no blue-shift for the equivalent LiTFSI-based LHCE.¹⁴

In conclusion, the differing impacts of TTE addition on anion coordination are in line with the diffusivities discussed above (see Figure 2f): the LiFSI-based LHCEs display strong ionic interactions, leading to a constant $D_{\text{anion}}/D_{\text{Li}}$ -ratio close to one. The LiTFSI-based electrolytes, however, show an increasing degree of ion dissociation upon dilution, leading to a larger fraction of free mobile anions. Since the coordination environment influences the dynamics of the species, the

enhanced ion dissociation can explain the observation of an increasing $D_{\text{anion}}/D_{\text{Li}}$ -ratio: as the self-diffusion coefficient is an average value over all anion species, the increasing fraction of free mobile anions leads to a stronger increase of D_{anion} compared to D_{Li} , resulting in the observed increase in the $D_{\text{anion}}/D_{\text{Li}}$ -ratio. This is probably due to the larger size of the TFSI^- anion, resulting in a better charge delocalization or the more amphiphilic character, which allows the anion to interact with the nonpolar diluent molecule at the internal interface.

Local Dynamics. Finally, our conclusions on the structure of the LHCE in dependence on the type of the anion are supported by measurements of spin–lattice (R_1) and spin–spin (R_2) relaxation rates of the cation and anion (^7Li and ^{19}F , resp.), as shown in Figure 5a,b, respectively. Both relaxation rates decrease with an increasing amount of diluents, in line with a decreasing viscosity, which implies enhanced local dynamics. This is also the reason, why for both ions R_1 and R_2 are the lowest in the lower concentrated FSI(1.8) LHCE, offering the highest overall system dynamics. Comparing the two electrolyte compositions at the saturation limit, we observe $\text{LiFSI}(1.2) > \text{LiTFSI}(1.8)$ for ^7Li relaxation rates, while the ^{19}F relaxation rates cannot be directly compared due to differing coupling constants and motional modes of the $-\text{CF}_3$ versus the $-\text{F}$ group. The ^7Li relaxation rate trend matches the already discussed strong ionic interactions and compact conducting salt phase leading to slower local Li^+ dynamics in the LiFSI(1.2) system in contrast to the LiTFSI(1.8) formulation.

For better comparability, Figure 5c depicts the R_2/R_1 relaxation rate ratios for both nuclei in all of the investigated systems. Analyzing these ratios has the following advantages: first of all, thereby it is possible to compare values obtained for different NMR-active nuclei since the gyromagnetic ratio, which is specific for each nucleus, is canceled. Moreover, this ratio is useful to identify the contribution of slow motions in concentrated electrolytes^{29,30} since according to theory R_1 reflects fast motions, while R_2 additionally contains contributions of slow motions. Thus, $R_2/R_1 = 1$ is expected for a very mobile, liquid system and R_2/R_1 is increasing with the local motional correlation time for constituents, which are not in motional narrowing.

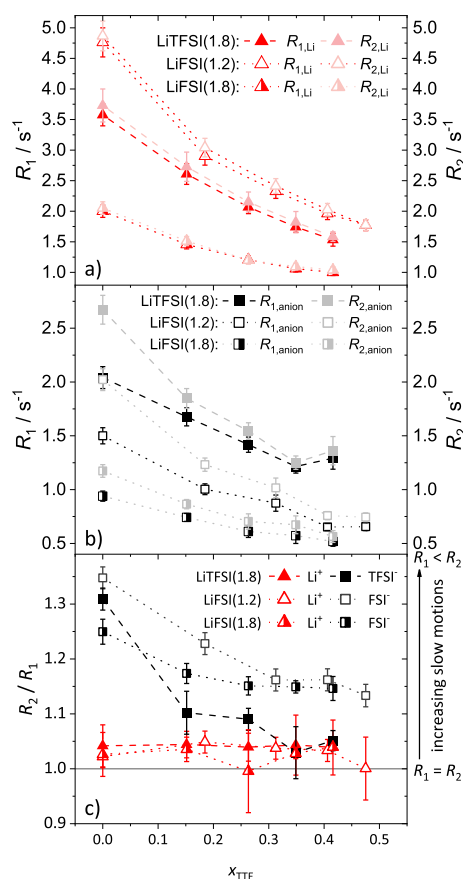


Figure 5. R_1 (dark red/black symbols) and R_2 (light red/gray symbols) relaxation rates of ⁷Li (a) and ¹⁹F (anion, b). (c) Ratio R_2/R_1 for both ions (⁷Li red triangles, ¹⁹F black squares). All three graphs are in dependence on x_{TTE} , and filled/open half-filled symbols correspond to LiTFSI(1.8), LiFSI(1.2), and LiFSI(1.8), rsp.

In the present case, $R_2/R_1 \approx 1$ is observed for the ⁷Li relaxation rate ratio of the Li^+ ions over the whole concentration range for all electrolyte formulations, while $R_2/R_1 > 1$ is found for the ¹⁹F relaxation rate ratio of all anions in the HCE system ($x_{TTE} = 0$). This difference is probably due to the fact that the Larmor frequency ω_0 of ¹⁹F is twice as large as $\omega_0(^{7}Li)$. Therefore, according to theory, ⁷Li nuclei are still in motional narrowing for larger correlation times τ_c compared to ¹⁹F (maximum of R_1 at $\tau_c = 0.616/\omega_0$). Thus, $R_2/R_1 \approx 1$ can be observed for ⁷Li, while $R_2/R_1 > 1$ is found for ¹⁹F, even though the dynamics and correlation times are similar.

However, the R_2/R_1 ¹⁹F relaxation rate ratio shows a strong decrease with increasing x_{TTE} approaching the values of Li^+ in the case of LiTFSI(1.8). Thus, dilution results in a much more flexible and locally dynamic TFSI[−] anion. This supports the view of weakly coordinated and thus rather mobile TFSI[−] anions stabilizing the TTE|conducting salt interface. Further, this beneficial interaction at the TTE|conducting salt interface might also lead to an enhanced flexibility of the interface, the formation of a larger surface area, and a higher mobility of the entire phases, resulting in this observed improved anion dynamics.

For the FSI[−] anion, Figure 5c reveals once more a deviating behavior: even though R_2/R_1 is decreasing, too, with the addition of TTE due to generally enhanced system dynamics, the ratio reaches a plateau value of $R_2/R_1 \approx 1.15$, which is still higher than one. Interestingly, this plateau is identical for

LiFSI(1.2) and LiFSI(1.8), while R_2/R_1 is different for these two formulations in the HCE and slightly diluted state, where we observe $LiFSI(1.2) > LiFSI(1.8)$. It seems that for low amounts of diluent, the factor dominating slow local motions is the different conducting salt concentration. Since it is lower in the LiFSI(1.8) formulation, the anion is less immobilized as compared to the more highly concentrated LiFSI(1.2) composition. In the case of strong dilution, the difference in behavior depending on the salt concentration vanishes. Due to entropic effects, we expect that more and more DME molecules partition from the conducting salt phase to the TTE phase when the dilution of the LHCE system increases. As already expected from the chemical shifts shown in Figures 3 and S3, we assume that more DME is dissolved in the TTE phase in the LiFSI(1.8) system compared to the higher concentrated LiFSI(1.2) system. Thus, toward a high amount of TTE, the conducting salt concentration in the conducting salt phases converges for both systems, leading to similar anion dynamics and equal R_2/R_1 ratios, as shown in Figure 5c.

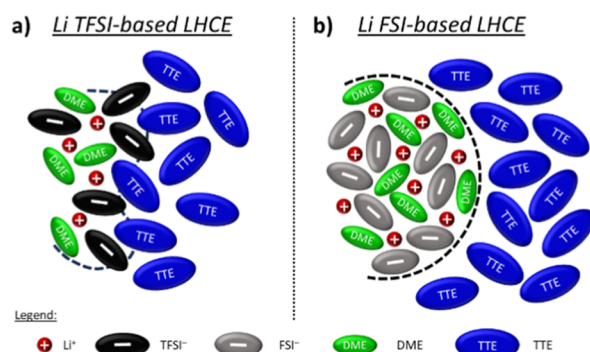
Once more, the observation of $LiTFSI(1.8) < LiFSI(1.8) \approx LiFSI(1.2)$ for the R_2/R_1 ratios of the anions in the stronger diluted region hints on a stronger salt aggregation in the LiFSI-based electrolytes and a less favorable interaction between FSI[−] and TTE at the TTE|conducting salt interface. In conjunction with this, the LiTFSI-based electrolyte might consist of smaller conducting salt-rich phases with a higher internal interface of TTE|conducting salt since TFSI[−] is more likely to interact with TTE molecules compared to FSI[−]. The tendency for creation of internal interface then presumably reduces the size and might alter the shape of the mesoscopic phase droplets, which enhances macroscopic transport and is in turn beneficial not only for anion but also for Li^+ macroscopic transport.

Overall Structural Image. In the following, we take all discussed findings into account in order to identify structural and dynamic differences of the LHCE in dependence on the anion employed. In the case of the LiFSI-based LHCE, dilution with TTE induces slightly enhanced ionic interactions (see blue-shift of the Raman band), and no enhancement of dissociation is observed. This implies that the coordination structure in the conducting salt phase remains intact. There is, however, a slight increase of the anion dynamics (reduction of R_2/R_1 , though motional narrowing is not reached, $R_2/R_1 > 1$). This can be attributed to an interfacial effect, where FSI[−] anions close to the TTE|conducting salt interface might exhibit slightly higher dynamics due to reduced Coulombic interactions. Furthermore, in line with unaltered ion coordination (see Raman results), we observed a constant self-diffusion coefficient ratio, D_{FSI}/D_{Li} upon dilution. This signifies that the transport mechanisms in the conducting salt phase are unaltered, and the self-diffusion enhancement due to dilution is identical for all species in the conducting salt phase. On the other hand, the ionic conductivity decreases, which reflects its dependence on the number density as the latter is decreasing with TTE addition.

In conclusion, upon dilution, the coordination in the conducting salt phase of LiFSI-based LHCEs remains largely unaltered, while a TTE|conducting salt interface is formed, causing only a minor enhancement of anion dynamics. A rough sketch of the situation is given in Scheme 2b.

For the LiTFSI-based LHCE, we found quite different properties upon dilution. First of all, an increased ion dissociation (larger share of free TFSI[−], see Raman data) is

Scheme 2. Schematic Description of the Interface between Conducting Salt-Rich and Diluent Phase in the LiTFSI-Based LHCE (a) and LiFSI-Based LHCE (b)



accompanied by an increased ionic conductivity as compared to the HCE. We attribute the increasing fraction of free anions to their enrichment at the growing TTE|conducting salt interface. The larger size and enhanced charge delocalization of the TFSI⁻ anion in comparison to FSI⁻ leads to a beneficial interaction with the unpolar TTE phase, which is why we assume that in this case the anion predominantly forms the interface toward the TTE phase (illustrated in Scheme 2a). Consequently, on average, TFSI⁻–Li⁺ coordination is reduced. This goes along with enhanced local dynamics of TFSI⁻ (see filled squares in Figure 5c) as well as an increased self-diffusion coefficient ratio $D_{\text{TFSI}}/D_{\text{Li}}$ (see filled squares in Figure 2f). Both observations are in strong contrast to the properties of the LiFSI-based LHCEs. In a similar FSI-based LHCE, Efav et al. even postulated an enrichment of DME at the interface,³¹ which underlines the opposing interfacial properties of these two anions. Further, we assume that the beneficial interaction of TFSI⁻ with TTE stabilizes the TTE|conducting salt interface and promotes its formation, resulting in flexible and dynamic phases in general.

Concerning an improvement in the ionic conductivity, the dilution of the conducting salt phase with TTE can be viewed as a competition of two effects: first, the addition of a nonionic phase reduces the overall charge carrier density, which reduces the ionic conductivity. Second, in the case of the LiTFSI-based LHCE, TTE addition promotes salt dissociation, enhancing ionic conductivity. In the case of the LiFSI-based LHCE, where salt dissociation is not promoted, the ionic conductivity decreases because of the dominance of the first effect. In contrast, for the LiTFSI-based LHCE, the enhanced salt dissociation dominates despite the existence of a neutral phase, causing an increase of the ionic conductivity. The ionic conductivity increase might potentially be further aided by enhanced mobility of the interfacial anions.

Thus, distinct differences in dependence on the type of anion are observed for LHCEs, resulting from a different interplay between the diluent and conducting salt-rich phase. As the interface is a critical part of LHCEs influencing their dynamics, these insights might help to make further improvements in the performance of this type of electrolyte.

CONCLUSIONS

In conclusion, the presented multimethod investigation of the structures and dynamics in a LiTFSI-based LHCE and two LiFSI-based systems elucidates the influences of the anion on the complex heterogeneous structure (salt-rich and diluent-rich

phases) of the LHCE and thus explains the observed differences in the dynamic properties. The comparison shows that a reduction of the viscosity by an increasing dilution with TTE leads to higher self-diffusion coefficients but not necessarily higher ionic conductivities. While σ slightly increases in case of the LiTFSI-based system compared to the HCE, this is not the case for both LiFSI-based LHCE formulations. Therefore, the addition of a diluent does not inevitably enhance the ionic conductivity of the resulting electrolyte. As shown here, this depends, for example, on the anion used in the LHCE.

First of all, the discussed structural properties (Raman shifts and ¹H NMR chemical shifts) do not change tremendously with an increasing amount of diluent, which is expected since the diluent should not affect the ion coordination and form an independent phase. Thus, the condition of structural heterogeneity seems to be fulfilled for the LHCE formulations investigated here. Nevertheless, by comparing the three slightly different systems and various properties, we were able to see some deviating behavior. We quantified the partitioning of DME into the diluent-rich phase and found it to be low and similar for all systems. Therefore, the anion structure and its local coordination appear to be crucial: on the one hand, upon dilution of the HCE with the diluent, TFSI⁻ anions seem to interact with TTE molecules at the growing TTE|conducting salt interfaces, leading to a higher degree of ion dissociation and a larger share of mobile anions. On the other hand, FSI⁻ anions appear to interact less likely with TTE molecules, resulting in stronger ionic interactions and largely immobilized anions. Considering the different sizes of the anions, the charge delocalization of the larger TFSI⁻ anion might be superior to the smaller FSI⁻ anion, leading to the beneficial interaction with the uncharged TTE molecules. Moreover, TFSI⁻ has a stronger amphiphilic character, which further stabilizes the interface toward the nonpolar diluent-rich phase. Due to this different interfacial behavior, it is likely that the size and flexibility of the entire conducting salt-rich phase also depend on the anion structure: while the interactions between TFSI⁻ and TTE might lead to generally quite flexible and mobile conducting salt-rich and TTE-rich phases, resulting in increased dynamics, LiFSI-based LHCEs seem to form a more compact and less dynamic conducting salt-rich phase.

All in all, from our results, we conclude that the diluent|conducting salt interface is a crucial factor influencing the structural and dynamic properties of the electrolyte. We have demonstrated the key role of the anion structure: in particular, strongly surface active anions will be beneficial in future optimized LHCEs.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.5c01566>.

Electrolyte compositions, densities, NMR spectra, and deconvolution of Raman spectra (PDF)

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Notes

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REFERENCES

- (1) Cao, X.; Jia, H.; Xu, W.; Zhang, J.-G. Review—Localized High-Concentration Electrolytes for Lithium Batteries. *J. Electrochem. Soc.* **2021**, *168* (1), 010522.
- (2) Yamada, Y.; Wang, J.; Ko, S.; Watanabe, E.; Yamada, A. Advances and issues in developing salt-concentrated battery electrolytes. *Nat. Energy* **2019**, *4* (4), 269–280.
- (3) Hu, J.; Ji, Y.; Zheng, G.; Huang, W.; Lin, Y.; Yang, L.; Pan, F. Influence of electrolyte structural evolution on battery applications: Cationic aggregation from dilute to high concentration. *Aggregate* **2022**, *3* (1), No. e153.
- (4) Suo, L.; Hu, Y. S.; Li, H.; Armand, M.; Chen, L. A new class of Solvent-in-Salt electrolyte for high-energy rechargeable metallic lithium batteries. *Nat. Commun.* **2013**, *4*, 1481.
- (5) Yamada, Y.; Yaegashi, M.; Abe, T.; Yamada, A. A super-concentrated ether electrolyte for fast-charging Li-ion batteries. *Chem. Commun.* **2013**, *49* (95), 11194–11196.
- (6) Ren, X.; Zou, L.; Cao, X.; Engelhard, M. H.; Liu, W.; Burton, S. D.; Lee, H.; Niu, C.; Matthews, B. E.; Zhu, Z.; et al. Enabling High-Voltage Lithium-Metal Batteries under Practical Conditions. *Joule* **2019**, *3* (7), 1662–1676.
- (7) Chen, S.; Zheng, J.; Mei, D.; Han, K. S.; Engelhard, M. H.; Zhao, W.; Xu, W.; Liu, J.; Zhang, J. G. High-Voltage Lithium-Metal Batteries Enabled by Localized High-Concentration Electrolytes. *Adv. Mater.* **2018**, *30* (21), No. e1706102.
- (8) Dokko, K.; Tachikawa, N.; Yamauchi, K.; Tsuchiya, M.; Yamazaki, A.; Takashima, E.; Park, J.-W.; Ueno, K.; Seki, S.; Serizawa, N.; et al. Solvate Ionic Liquid Electrolyte for Li-S Batteries. *J. Electrochem. Soc.* **2013**, *160* (8), A1304–A1310.
- (9) Sudoh, T.; Ikeda, S.; Shigenobu, K.; Tsuzuki, S.; Dokko, K.; Watanabe, M.; Shinoda, W.; Ueno, K. Li-Ion Transport and Solution Structure in Sulfolane-Based Localized High-Concentration Electrolytes. *J. Phys. Chem. C* **2023**, *127* (25), 12295–12303.
- (10) Watanabe, Y.; Ugata, Y.; Ueno, K.; Watanabe, M.; Dokko, K. Does Li-ion transport occur rapidly in localized high-concentration electrolytes? *Phys. Chem. Chem. Phys.* **2023**, *25* (4), 3092–3099.
- (11) Bergstrom, H. K.; McCloskey, B. D. Ion Transport in (Localized) High Concentration Electrolytes for Li-Based Batteries. *ACS Energy Lett.* **2024**, *9* (2), 373–380.
- (12) Dokko, K.; Watanabe, D.; Ugata, Y.; Thomas, M. L.; Tsuzuki, S.; Shinoda, W.; Hashimoto, K.; Ueno, K.; Umebayashi, Y.; Watanabe, M. Direct Evidence for Li Ion Hopping Conduction in Highly Concentrated Sulfolane-Based Liquid Electrolytes. *J. Phys. Chem. B* **2018**, *122* (47), 10736–10745.
- (13) Zhang, H.; Lin, Y.; Wang, J. Design of Localized High-Concentration Electrolytes from the Perspective of Physicochemical Properties. *J. Phys. Chem. Lett.* **2024**, *15* (32), 8378–8386.
- (14) Liu, J.; Kaneko, T.; Ock, J.-y.; Kondou, S.; Ueno, K.; Dokko, K.; Sodeyama, K.; Watanabe, M. Distinct Differences in Li-Deposition/Dissolution Reversibility in Sulfolane-Based Electrolytes Depending on Li-Salt Species and Their Solvation Structures. *J. Phys. Chem. C* **2023**, *127* (12), 5689–5701.
- (15) van Ekeren, W.; Hall, A.; Lahtinen, K.; Younesi, R. The Solvation Structure of Localized High Concentration Electrolytes. *ChemElectroChem* **2024**, *11* (11), No. e202400050.
- (16) Hahn, E. L. Spin echoes. *Phys. Rev.* **1950**, *80* (4), 580–594.
- (17) Vold, R. L.; Waugh, J. S.; Klein, M. P.; Phelps, D. E. Measurement of Spin Relaxation in Complex Systems. *J. Chem. Phys.* **1968**, *48* (8), 3831–3832.
- (18) Carr, H. Y.; Purcell, E. M. Effects of Diffusion on Free Precession in Nuclear Magnetic Resonance Experiments. *Phys. Rev.* **1954**, *94* (3), 630–638.
- (19) Meiboom, S.; Gill, D. Modified Spin-Echo Method for Measuring Nuclear Relaxation Times. *Rev. Sci. Instrum.* **1958**, *29* (8), 688–691.
- (20) Wiemhöfer, H. D.; Grünebaum, M.; Hiller, M. M., Micro electrode liquid measurement cell, WIPO Utility Patent, No. WO 2014139494 A1 2014.
- (21) Lin, S.; Hua, H.; Li, Z.; Zhao, J. Functional Localized High-Concentration Ether-Based Electrolyte for Stabilizing High-Voltage Lithium-Metal Battery. *ACS Appl. Mater. Interfaces* **2020**, *12* (30), 33710–33718.
- (22) Nakanishi, A.; Ueno, K.; Watanabe, D.; Ugata, Y.; Matsumae, Y.; Liu, J.; Thomas, M. L.; Dokko, K.; Watanabe, M. Sulfolane-Based Highly Concentrated Electrolytes of Lithium Bis-(trifluoromethanesulfonyl)amide: Ionic Transport, Li-Ion Coordination, and Li-S Battery Performance. *J. Phys. Chem. C* **2019**, *123* (23), 14229–14238.
- (23) Su, L.; Lu, F.; Liu, X.; Wang, C.; Gao, Y.; Passerini, S.; Zheng, L.; Xinpei, G. Molecular insight into nano-heterogeneity of localized high-concentration electrolyte: Correlation with lithium dynamics and solid-electrolyte interphase formation. *J. Power Sources* **2023**, *557*, 232545.
- (24) Wang, L.; Luo, Z.; Xu, H.; Piao, N.; Chen, Z.; Tian, G.; He, X. Anion effects on the solvation structure and properties of imide lithium salt-based electrolytes. *RSC Adv.* **2019**, *9* (71), 41837–41846.
- (25) Hockmann, A.; Ackermann, F.; Diddens, D.; Cekic-Laskovic, I.; Schönhoff, M. Heterogeneous Li coordination in Solvent-in-Salt electrolytes enables high Li transference numbers. *Faraday Discuss.* **2024**, *253*, 343–364.

(26) Seo, D. M.; Borodin, O.; Han, S.-D.; Boyle, P. D.; Henderson, W. A. Electrolyte Solvation and Ionic Association II. Acetonitrile-Lithium Salt Mixtures: Highly Dissociated Salts. *J. Electrochem. Soc.* **2012**, *159* (9), A1489–A1500.

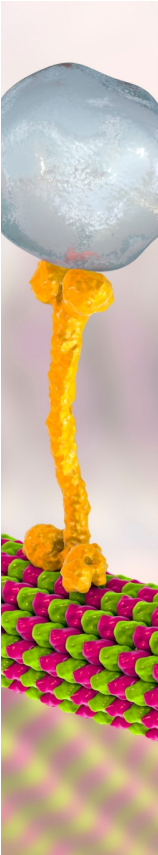
(27) Zhou, Q.; Boyle, P. D.; Malpezzi, L.; Mele, A.; Shin, J.-H.; Passerini, S.; Henderson, W. A. Phase Behavior of Ionic Liquid-LiX Mixtures: Pyrrolidinium Cations and TFSI⁻ Anions - Linking Structure to Transport Properties. *Chem. Mater.* **2011**, *23* (19), 4331–4337.

(28) Han, S.-D.; Borodin, O.; Seo, D. M.; Zhou, Z.-B.; Henderson, W. A. Electrolyte Solvation and Ionic Association. *J. Electrochem. Soc.* **2014**, *161* (14), A2042–A2053.

(29) Kunze, M.; Paillard, E.; Jeong, S.; Appetecchi, G. B.; Schönhoff, M.; Winter, M.; Passerini, S. Inhibition of Self-Aggregation in Ionic Liquid Electrolytes for High-Energy Electrochemical Devices. *J. Phys. Chem. C* **2011**, *115* (39), 19431–19436.

(30) Kunze, M.; Jeong, S.; Paillard, E.; Schönhoff, M.; Winter, M.; Passerini, S. New Insights to Self-Aggregation in Ionic Liquid Electrolytes for High-Energy Electrochemical Devices. *Adv. Energy Mater.* **2011**, *1* (2), 274–281.

(31) Efaw, C. M.; Wu, Q.; Gao, N.; Zhang, Y.; Zhu, H.; Gering, K.; Hurley, M. F.; Xiong, H.; Hu, E.; Cao, X.; et al. Localized high-concentration electrolytes get more localized through micelle-like structures. *Nat. Mater.* **2023**, *22* (12), 1531–1539.



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